
EE 570: Advanced Communication Theory

Course Code	EE 570
Course Name	Advanced Communication Theory
L-T-P-C	3-0-2-4
Prerequisites	IC210: Probability, Statistics and Random Processes, EE 304: Communication Theory or the instructor's consent
Intended for	BTech/MTech/MTech (R)/ PhD students of the SCEE.
Mutual Exclusion	None so far
Approval	54th BoA (May 2024)

Course Contents

- **Digital modulation schemes:** Bandpass and lowpass signal representation, Digital modulation schemes (PAM, PM, QAM, Multidimensional Signals, CPFSK, CPM) and their corresponding optimal receivers and error probabilities for AWGN Channel. (12 Hours)
- **Carrier and Symbol Synchronization:** Importance in signal demodulation, carrier frequency and phase estimation – decision directed and power of N methods, timing estimation - spectral-line, MMSE, and ML methods, joint carrier and symbol synchronization. (8 Hours)
- **Equalization:** Optimal zero-forcing equalization, Linear, Decision-feedback, Adaptive Linear, Adaptive Decision-feedback, and Blind equalization. (12 Hours)
- **Multichannel and Multicarrier Systems:** Wireless & AWGN multi-channels, Multicarrier communications: OFDM – modulation and demodulation, spectral characteristics, bit and power allocation, channel. (8 Hours)
- **Case Studies:** A brief overview of modern communication/broadcast technologies. (3 Hours)

Laboratory Modules

- Not envisaged

Text Books

1. J. G. Proakis and M. Salehi, **Digital Communications**, 5th Edition, McGraw-Hill, Prentice Hall, 2007.
2. R. G. Gallager, **Principles of Digital Communication**, Cambridge Univ. Press, 2008.

References

1. B. Sklar, **Digital Communications: Fundamentals and Applications**, 2nd Edition, Prentice Hall, 2001.
2. John R. Barry, David G. Messerschmitt, and Edward A. Lee, **Digital Communication**, 3rd Edition, Kluwer Academic Publishers, 2003.
3. A. Lapidoth, **A Foundation in Digital Communication**, Cambridge Univ. Press, 2009.
4. Simon Haykin, **Digital Communications**, Wiley Publishing, 2006.